

SET-2



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Department of Computer Science & Engineering

Ist Lab Internal - Set 2

Subject Name: Software Engineering Subject Code: 23CC505PC

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Faculty: K Yasaswini Lab: FS2

In today's digital world, managing ticket bookings manually can lead to long queues, delays, and operational inefficiencies. To solve this, your development team has been assigned to build a web-based **E-Ticketing System** that simplifies the process of booking tickets for movies, concerts, and public events.

1. **Clone the Online Ticketing System repository to your eclipse software from the URL below and execute the project by changing the required changes in the POM.xml file and make it a successful build. [40 Marks]**

<https://github.com/ssvkotamraju/E-ticketing.git>

- a. You have joined a development team working on an online ticketing application. Your team lead provides the GitHub repository URL. Explain the steps required to clone the repository and import the project into Eclipse.
- b. After importing the E-Ticketing project into Eclipse, several dependency-related errors are displayed. How would you identify and resolve the Maven dependency problems?
- c. The project fails to compile because the Java compiler configuration in pom.xml does not match the Java version installed on your system. How would you troubleshoot and correct the configuration?
- d. You execute mvn clean install, but the build fails during the compilation phase. Explain how you would use the Maven error output to identify the root cause.
- e. The E-Ticketing application needs to be deployed as a web application, but Maven generates an unexpected package type. Which section of pom.xml would you inspect and modify?
- f. After modifying pom.xml, the application behaves differently from the previous build. Explain why executing a clean Maven build can help and which Maven command you would use.

- g. Maven displays BUILD SUCCESS, but the E-Ticketing application does not open correctly in the browser. What additional checks would you perform to verify that the application is running correctly?
- h. The development team wants a quick production package because all tests were already executed successfully in the CI pipeline. Which Maven command could be used to avoid running tests during packaging?
- i. The build fails because Maven cannot download a dependency. How would you determine whether the problem is caused by the dependency declaration, repository configuration, or network connectivity?
- j. You have fixed the ticket-booking calculation logic and want Maven to execute the project's tests. Which Maven command would you use?

2. Working with Version Control (GitHub, GitLab, etc.) [40 Marks]

You're working as a junior developer on a railway ticket booking portal. Solve the following scenarios for the application.

- a. You receive the railway ticket-booking project as a local folder that is not under Git version control. How would you create the first commit?
- b. The railway portal needs a new feature that allows users to search trains based on source and destination. Explain how you would create a separate branch and develop this feature without directly modifying main.
- c. You have modified the ticket-booking page and added a new CSS file. You want to review the changes before committing them. Which Git commands would you use?
- d. You accidentally modify the train-search page and want to discard the changes because they have not yet been committed. Which Git command would you use?
- e. You have created several local commits while experimenting with the ticket-booking functionality. None of the commits have been pushed to GitHub. You want to move your branch back to an earlier commit.
- f. Several developers have created temporary branches for completed features. The branches are no longer required after their changes have been merged. How would you identify and delete the unused branches?
- g. Developer A implements train-search functionality while Developer B implements passenger-registration functionality. Both features are developed on separate branches. Explain how you would merge them into the main development branch.
- h. A teammate cannot directly push to your repository but sends you their changes as a Git patch. Explain how you would create, inspect, and apply the patch to your local project.
- i. You receive a patch from a teammate, but the patch cannot be applied cleanly because your local version contains conflicting changes. How would you troubleshoot and resolve the problem?
- j. The railway ticket-booking application has been tested successfully. You now need to upload the project and its Git history to your GitHub repository. Explain how you would configure the remote repository and push the required branch.

3. Run the Project with Docker [20 Marks]

- i. Create a Dockerfile and build a Docker image named OTS to push it to DockerHUB.
- ii. Run the image in a container and expose port 8080. Check if the app runs at <http://localhost:8080>.

- a) You execute docker build, but Docker reports an error while creating the OTS image. What steps would you follow to identify and correct the problem?
- b) The E-Ticketing application listens on port 8080 inside the container, but localhost:8080 cannot access it. Explain how Docker port mapping should be configured.
- c) The OTS Docker image builds successfully, but the container stops immediately after it starts. What commands or logs would you inspect to determine the cause?
- d) The E-Ticketing application works on one developer's computer but fails on another because of differences in Java and server configurations. Explain how Docker can solve this problem.
- e) The E-Ticketing application generates a .war file after running Maven. You need to deploy this WAR file inside a Tomcat-based Docker container and make the application available on port 8080. What changes would you make to the Dockerfile?